

Nikhilesh Vijay Chirutala

AI Software Engineer | Python Backend Engineer

+91-9618873598 • nikhilesh.chirutala@gmail.com • [Linkedin](#) • [Github](#) • [Portfolio](#) • Bengaluru, IN

SUMMARY

Python Software Engineer with 3 years of experience building production applications, backend services, API integrations, and workflow automation across AI, CRM, and cloud contact-center platforms. Hands-on experience integrating LLM-powered capabilities, building a RAG-based chatbot, and developing AI-enabled workflows and system integrations using Python and FastAPI. Experienced in REST APIs, AWS, automation workflows, data processing, production troubleshooting, and deploying reliable software solutions. Strong problem-solving mindset with experience translating business requirements into practical automation and AI-assisted solutions.

TECHNICAL SKILLS

Programming: Python, JavaScript (ES6+), TypeScript, SQL

AI & LLM: LLM Integration, RAG, AI Automation, Prompt Engineering, AI-Powered Workflows, MCP Integration

Backend & APIs: FastAPI, REST APIs, API Development, Backend Automation, API Integration, WebSockets

Automation & Workflows: Workflow Automation, Process Automation, Task Workflows, Data Processing, Scheduled Workflows, AI-Assisted Workflows

Cloud & DevOps: AWS, Amazon S3, AWS Lambda, Vercel, GitHub Actions, CI/CD, Linux

Databases: MongoDB, PostgreSQL, MySQL, Neon DB, SQL

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Jinja2

Integrations & Tools: Twilio, GitHub API, Postman, Git, GitHub, Datadog, Grafana, Sentry

WORK EXPERIENCE

Software Engineer • [CA One Tech Cloud Inc.](#)

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services for a high-volume cloud contact-center platform, supporting automated call-recording processing, data workflows, storage operations, and production systems.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, implementing automated processing, feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, processing limits, retries, response validation, and result persistence.
- Developed automated validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Supported production deployment and rollout of Python automation services across multiple clusters, including configuration, validation, monitoring, troubleshooting, and deployment verification.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage processes.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry to monitor service health, track errors, diagnose processing failures, and support incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing root-cause analysis and implementing fixes to maintain reliable automated processing workflows.

Software Engineer • [Mantra Technologies](#)

Oct 2023 – Mar 2026 | Hyderabad, India

- Delivered production features and backend workflows across multiple applications, translating business requirements into application modules, API integrations, automation workflows, and end-to-end functionality.
- Developed Python and FastAPI services and REST APIs supporting business processes, external integrations, data flows, and AI-enabled application capabilities.
- Implemented authentication, authorization, external service integrations, validation, and workflow logic while troubleshooting integration and application issues.
- Supported production releases through testing, debugging, root-cause analysis, deployment validation, and defect resolution.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts into a unified application.
- Implemented portfolio workflows for tracking positions, stocks, and mutual funds, along with reports and investment-level information across aggregated accounts.

- Developed stock and mutual fund screeners with configurable filtering workflows and detailed asset pages for analyzing investments and underlying information.
- Designed and implemented a backtesting workflow for evaluating trading strategies against historical market data and analyzing strategy performance.
- Built a Retrieval-Augmented Generation (RAG) chatbot that combines portfolio and financial information with LLM capabilities to provide contextual insights and answer portfolio-related user queries.
- Developed Python and FastAPI backend APIs supporting portfolio aggregation, investment analysis, screening, backtesting, and AI-powered workflows.

CRM.io — AI-Driven CRM & Automation Platform

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM

- Developed Python and FastAPI backend services and REST APIs for an AI-driven CRM, implementing business workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built a real-time CRM dialer integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented task cadence workflows that allow users to create sequences of tasks as a single process and progress through scheduled follow-ups, with Firebase Cloud Messaging used for notifications.
- Built document management workflows using Amazon S3, including browser-to-S3 uploads and automated association of documents with CRM records.
- Implemented a CRM-wide soft-delete and retention workflow using AWS Lambda to automate permanent removal of records after the retention period.
- Built account-level reporting dashboards and an MCP integration exposing CRM capabilities as tools, enabling an AI bot to retrieve CRM information and trigger supported operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur